

VITTORIO CENTORE

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Education

University of Michigan

B.S. in Computer Science

Ann Arbor, MI

Expected May 2027

- **GPA:** 3.71/4.00
- **Coursework:** Data Structures & Algorithms, Web Systems, Object Oriented Programming, Computer Organization, Foundations of Computer Science, Discrete Mathematics, Linear Algebra, Intro to Data Science
- **Activities:** Atlas Digital Consulting Group, Reach Consulting Group, Michigan Rover Team, Italian Association

Experience

AgentMail (YC S25)

Ann Arbor, MI

Software Engineering Analyst

Sep 2025 – Present

- Engineered an AI sales agent via OpenAI's Agents SDK and AgentMail's API, sustaining a continuously replenished pool of AI-generated emails with sub-8-second initial availability, reducing manual client outreach efforts by 60%.
- Architected a full-stack, interactive React demo that visualized real API calls through a narrative walkthrough, guiding users through how AI agents use AgentMail's Inbox API and boosting product engagement by 35%.

'Sup | Seed Round Startup

Miami, FL

Software Engineering Intern

Jun 2025 - Sep 2025

- Developed a mock interview tool with Next.js and FastAPI, integrating Gemini's API to analyze user responses without speech-to-text conversion, enabling faster, more precise feedback and improving practice efficiency by 40%.
- Deployed facial tracking with MediaPipe and face-api.js to derive emotion-based confidence signals, feeding them into an LLM-based grading system to deliver more personalized feedback and improve evaluation accuracy by 60%.
- Constructed a high-performance media compression pipeline with FFmpeg that reduced audio/video storage by 80%

U-M School of Information

Ann Arbor, MI

Software Engineer

Feb 2025 – May 2025

- Designed a scalable backend pipeline with RESTful endpoints to filter and rank award-eligible faculty in real time, reliably processing 8,000 profiles in ~13 seconds through a React-based chat interface optimized for low latency
- Optimized backend data handling for large-scale faculty datasets, using Pandas DataFrames to pre-filter records and PyTorch to speed up model execution while handling higher request volume, reducing processing time by 30%
- Productionized backend and model inference on remote GPU servers, using automated CI/CD pipelines with unit tests for faculty ranking logic and integration tests for chat request handling, improving correctness by 20%

Michigan Rover Team

Ann Arbor, MI

Fullstack Developer

Sep 2024 – May 2025

- Rebuilt the camera dashboard to dynamically switch mission-specific camera feeds using Vue.js, TypeScript, and Bootstrap, enabling real-time switching across live camera streams and contributing to a top 10% national finish
- Improved robotic arm usability by implementing selectable manual and inverse kinematics modes, allowing operators to choose between precise or coordinated arm movements, speeding up mission time by 20% from previous year

Projects

Rush (Student Org Platform)

- Programmed a student org recruiting tracker that centralizes applications, deadlines, and events across 800+ UMich clubs, enabling students to track per-club status, keep notes, browse a discovery feed, and receive email reminders.
- Integrated a Firestore backend with Redis caching for hot pages (feed/weekly) and scheduled jobs for email reminders, reducing feed load time by 40% while ensuring concurrency-safe updates across events and status tracking.

Transaction Risk Engine

- Trained a PyTorch model to detect fraud in credit card transactions using computed risk signals (spending velocity, merchant history, and location), deployed as a FastAPI service backed by PostgreSQL, achieving 77% recall.

Distributed MapReduce Framework

- Implemented Python-based distributed MapReduce framework with a multi-threaded architecture, custom manager and worker classes, and network (TCP/UDP) communication, resulting in fault tolerant, efficient data processing.

Technical Skills

Languages: Python, C/C++, Java, TypeScript, JavaScript, HTML/CSS, SQL

Technologies: React.js, Next.js, Tailwind, FastAPI, AWS, Firebase, PostgreSQL, Git, Docker

Interests: Soccer (Napoli Supporter), Cars, Watches, Snowboarding, Marvel, Travel, Music